ECA0806-Analog & Digital Communication

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Experiment-3

MATLAB-BASED MODELLING OF VSB MODULATOR AND DEMODULATOR FOR ANALOG TELEVISION COMMUNICATION APPLICATIONS

Program (Matlab Code):

```
% Experiment 3: VSB Modulator and Demodulator (Toolbox-Free Version)
clc;
clear;
close all;

% Simulation Parameters
fs = 100000;          % Sampling Frequency (Hz)
t = 0:1/fs:0.02;      % Time Vector (s)
fm = 500;             % Message Frequency (Hz)
fc = 5000;            % Carrier Frequency (Hz)
% Message Signal
m = cos(2 * pi * fm * t);
% Carrier Signal
c = cos(2 * pi * fc * t);
% DSB-SC Modulation
dsb = m .* c;

% VSB Filter via Frequency Domain (Bypasses Signal Processing Toolbox)
N = length(dsb);
X = fftshift(fft(dsb));
f = linspace(-fs/2, fs/2, N);

% Create a VSB filter mask shaping the sidebands around fc
H = ones(size(f));
transition_idx = (abs(f) >= fc) & (abs(f) <= fc + 1000);
H(transition_idx) = linspace(1, 0, sum(transition_idx));
H(abs(f) > fc + 1000) = 0;

% Apply filter
X_filtered = X .* H;
vsb = real(ifft(ifftshift(X_filtered)));

% Coherent Demodulation
demod = vsb .* (2 * cos(2 * pi * fc * t));

% Low-Pass Filter for Message Recovery (Moving Average Filter)
window_size = 40;
b_lpf = ones(1, window_size) / window_size;
rec = filter(b_lpf, 1, demod);
rec = rec / max(abs(rec)) * max(abs(m)); % Scale amplitude for clarity
```

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```
% Display Results in Time Domain
figure('Name', 'VSB Modulation and Demodulation', 'NumberTitle', 'off');

subplot(5, 1, 1);
plot(t, m, 'LineWidth', 1.2);
title('Message Signal');
grid on;

subplot(5, 1, 2);
plot(t, c, 'LineWidth', 1.2);
title('Carrier Signal');
grid on;

subplot(5, 1, 3);
plot(t, dsb, 'LineWidth', 1.2);
title('DSB-SC Signal');
grid on;

subplot(5, 1, 4);
plot(t, vsb, 'LineWidth', 1.2);
title('VSB Signal');
grid on;

subplot(5, 1, 5);
plot(t, rec, 'r', 'LineWidth', 1.5);
title('Recovered Message Signal');
grid on;
```

Output Waveform :

